Where Does FireSET's Fire Data Come From?

A cross-source audit of ignition points, perimeters, timing, size and cause for 586 California wildfires, 2018–2025

fire-data-revalidate
https://github.com/utkrshkmr/fire-data-revalid

Run of 13 October 2026

Abstract

FireSET is a 586-fire California training set whose locations and times are taken from NIFC's WFIGS (IRWIN) feed and whose perimeters come from CAL FIRE FRAP. We asked how far those values can be corroborated by every other U.S. federal and California agency source that reports the same fires, and by three satellite products. We built and ran `fire-source-audit` (`fsa`), a probe → fetch → match → consensus pipeline over 13 agency sources and 3 satellite products. Three results stand out. (1) *The sources are far less independent than their catalog suggests*: FPA-FOD, InFORM, FireStat and MTBS largely re-publish the IRWIN record, so for a third of the fires there is no second agency location at all. (2) *Where independent points exist they disagree by about 800 m*, and by more than 2 km on one fire in six. FireSET's point is within 100 m of the best-supported agency point on 435 fires; 62 fires have a point that lies outside the final perimeter while another agency point lies inside it. (3) *Discovery times carry systematic clock errors*: USFS FireStat runs 2 h ahead of Pacific time in 2018–19, and the USFS web service re-labels those values as UTC, producing −5 h and −7 h offsets that propagate into InFORM and FPA-FOD. GOES-R's first detection falls a median 10 minutes after FireSET's start time, which independently confirms FireSET's timing. Seven defects in the source catalog had to be fixed before any of these conclusions could be drawn; they are documented so the audit can be re-run.

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1 Question and scope

FireSET (*VIIRS-Qualified California Wildfires, 2018–2025*, finalised 2026-08-30) holds one row per fire for 586 fires: 538 individual fires and 48 components of complexes. Its `Ignition_Lat/Lon` are IRWIN incident points obtained through NIFC WFIGS, `Start_Local` is IRWIN's discovery time (`FireDiscoveryDateTime`), and the perimeter and `GIS_ACRES` come from CAL FIRE FRAP. The dataset documentation states that no row has been hand-edited.

The brief for this audit was to *corroborate and understand* those values against every other government source that reports them: where did each fire start, what was its final perimeter, when was it discovered and contained, how big was it, and what caused it. A secondary question was raised explicitly: are FPA-FOD, CAL FIRE and IRWIN really independent of one another, or do they copy each other?

What counts as a source. Only records published by a U.S. federal agency or the State of California, plus physical satellite measurements from NASA and NOAA. Academic compilations (ICS-209-PLUS), commercial feeds and community mirrors were excluded by design. Table 1 lists what was used.

Table 1: Sources used in the run. "Lineage" is the upstream system whose record the source carries; sources with the same lineage are counted once in any consensus. Tier A = record of the responsible agency or an independent physical measurement; B = official but operational or second-hand; C = derivative merge.

Source	Agency	Gives	Lineage	Tier
WFIGS Incident Locations	NIFC / IRWIN	point, times, size, cause	IRWIN	A
InFORM Fire Occurrence (public)	NIFC	final-report point, times, size, cause	per record†	A
CAL FIRE incident list	CAL FIRE	map point, start time, acres	CALFIRE-OPS	A
USFS EDW FireStat points	USDA Forest Service	point, times, size, cause	USFS-FIRESTAT	A
FPA-FOD 7th ed. (1992–2024)	USFS Research	point, date+time, size, cause	per record†	A
MTBS fire-occurrence points	USGS / USFS GTAC	reported point (in `Event_ID`), date, acres	MTBS-REPORTED	B
CAL FIRE FRAP perimeters	CAL FIRE	final perimeter, dates, acres, cause	CALFIRE-FRAP	A
NIFC perimeter history (to 2021)	NIFC	perimeter, acres	from `SOURCE`	A
WFIGS Interagency Perimeters	NIFC	latest perimeter, acres	NIFC-WFIGS	A
USFS EDW final perimeters	USDA Forest Service	perimeter, acres	USFS-FIRESTAT	A
MTBS burned-area boundaries	USGS / USFS GTAC	Landsat dNBR perimeter, acres	MTBS	A
USGS combined fire polygons	USGS	merged perimeter (to 2020)	USGS-COMBINED	C
NASA FIRMS VIIRS 375 m	NASA	active-fire detections (S-NPP, NOAA-20)	SAT-LEO	A
NOAA GOES-R ABI FDC	NOAA	5-minute fire detections, 2 km	SAT-GEO	A
NOAA HMS fire points	NOAA NESDIS	analyst-checked daily points	SAT-HMS	B

†Compiled sources take the lineage of the system each record was copied from (§4).

2 Method

2.1 Pipeline

Figure 1 shows the five stages. Every stage is resumable and every raw pull is cached, so the analysis can be re-run offline. The full 586-fire run takes about 100 seconds on 16 cores once the cache is populated; fetching takes 15–30 minutes of polite (0.3 s-spaced) requests plus about 20 minutes for the optional GOES-R scan.

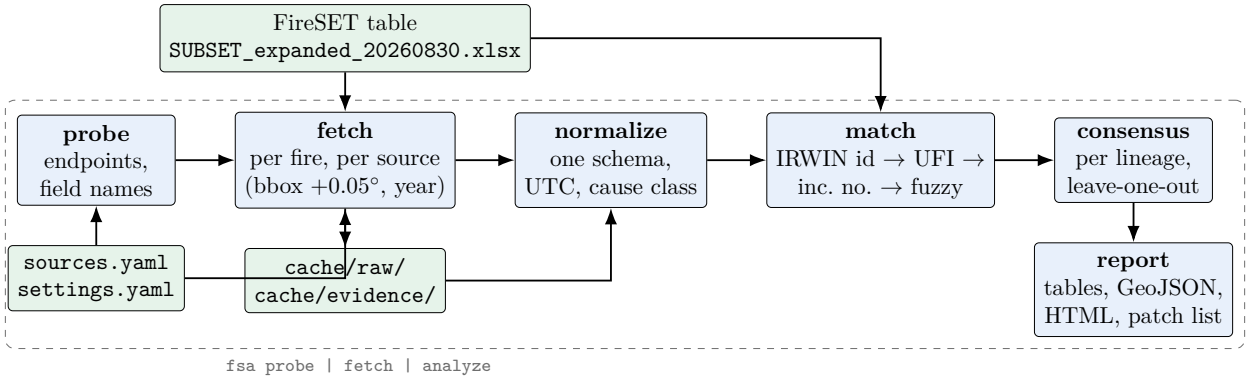


Figure 1: The `fsa` pipeline. Fetch runs where the agency hosts are reachable (a login node); match, consensus and report run anywhere from the cache.

2.2 Matching

For each fire, candidate records within the FireSET bounding box (buffered 0.05°) and the fire's season are scored. Identifier evidence wins outright: the same IRWIN GUID scores 1.00, the same `UniqueFireIdentifier` 0.97, the same incident number in the same year with a consistent location 0.90. Otherwise a fuzzy score combines name similarity (0.45), date proximity (0.25), place (0.20) and size (0.10), capped at 0.89 and halved when the names differ. A record is accepted at 0.60, and flagged ambiguous when a different record is within 0.05 of it. Two guards were added during the run: a fuzzy match more than 15 days from the FireSET start, or a perimeter dated before the fire or over a year after it, is rejected as another fire with the same name (21 such matches existed before the guard, e.g. CAL FIRE's "Peak Fire" of 2024 for Peak 2023).

2.3 Lineages and copy detection

Sources that re-publish one upstream record must not be allowed to "agree" with each other. The catalog assigns each source a lineage, but the run showed that is not enough (§4), so two mechanisms were added. First, compiled sources take their lineage *per record*: FPA-FOD from its `SOURCE_SYSTEM` column, InFORM from `CreatedBySystem`. Second, for each fire, two reported points from different lineages are treated as one report when they are within 5 m (float32 storage and Web Mercator round trips move a copied point by 1–2 m), or when one is exactly the other rounded to three or four decimals (the MTBS `Event_ID` encodes thousandths of a degree). Their lineages are merged for that fire before any consensus is built. Every detected copy is written to `identical_point_pairs.csv`.

2.4 Consensus and per-source error

The ignition consensus is the trust-weighted geometric median (Weiszfeld) of one medoid point per lineage. The agreement class is set by the largest distance between any two lineages: *agree* ≤ 500 m, *minor* ≤ 2 km, *major* beyond, or *single lineage*. The perimeter consensus rasterises one polygon per lineage at 30 m and keeps cells covered by at least half the lineages (with two lineages,

their union). Every per-source error compares the source with the consensus of the *other* lineages, so a source never grades itself; with only two lineages that disagree by more than 2 km, no reference is formed.

2.5 Satellite evidence

VIIRS detections inside the consensus perimeter (+1 km) give the first-overpass time and centre; detections more than 12 h before the earliest reported discovery are ignored, since they are almost always a neighbouring or earlier fire. GOES-R ABI fire-detection files (every 5 minutes) are scanned from 6 h before to 3 h after the reported discovery within 5 km of the FireSET point; the earliest fire-mask pixel is the tightest independent bound on ignition time. NOAA HMS analyst points are kept apart from VIIRS because they mix coarse GOES and AVHRR pixels with analyst timing.

Whole-hour offsets of 2, 5, 7 or 8 h between a source's discovery time and the other lineages are suspected clock or time-zone errors. Both sides of such a pair look offset, so the satellite's first heat (GOES, else HMS, else VIIRS) acts as referee: the side farther from it is flagged.

3 The catalog had to be corrected first

The delivered source catalog had been checked against live service *metadata*; the run checked it against live *data*. Seven defects would each have produced false disagreements. Table 2 lists them; all are fixed in `config/sources.yaml` and the adapters, with regression tests.

Table 2: Defects found by running against live data, and what the audit would have reported without the fix.

Source	Problem	Effect if left alone
WFIGS (IRWIN)	The config compared `InitialLatitude/Longitude`. FireSET uses the feature geometry (IRWIN's *current* point of origin): it matches the geometry on 478 of 479 fires. The first-report point differs from it by > 500 m on 193 fires and by > 5 km on 56.	FireSET "disagrees" with its own source on a third of fires.
InFORM	No `IrwinID` or text cause fields; IDs are in `RelatedIrwinIDs`, cause is an integer `FireCauseID`. `FinalAcres` is null for every California record. California 2018–19 records are FireStat/WFMI back-fill, not IRWIN.	No ID matches, no cause, no size; wrong lineage.
CAL FIRE incidents	API moved to `incidents.fire.ca.gov` (old host returns HTTP 403 to non-generic user agents); needs `&year=` or only the current season is returned; `Started` is UTC, not Pacific; 294 records carry a bulk "extinguished" date of 2019-01-04.	Source unreachable.
FPA-FOD 7th ed.	No `IRWIN_ID` or `UNIQUE_FIRE_ID` columns. Records are copies of other systems (`SOURCE_SYSTEM`): 2020 = IA-IRWIN, 2021–24 = IA-INFORM.	Counted as independent of FireSET when it is the same IRWIN record.
MTBS points	`BurnBndLat/Lon` is the burned-area centroid (Camp 2018: 10 km from the origin). The reported origin is encoded in `Event_ID`.	Every large fire a "major disagreement".
FRAP, FPA-FOD, MTBS dates	Calendar dates stored as 00:00 UTC were shifted into the previous Pacific day.	Every alarm date one day early.
CAL FIRE FIRIS view	Holds only the current season (476 polygons, 2026).	False same-name matches; now disabled.

Two engineering problems also surfaced only at full scale. Raster-traced perimeters (MTBS, FIRIS) carry tens of thousands of vertices, and the exact boundary Hausdorff distance is $O(nm)$: the Dixie Fire alone took 222 s, and buffering its perimeter for the detection zone took a further minute. Simplifying boundaries to 10 m (or $\sqrt{\text{area}}/500$ for very large fires) before the distance, and buffering a simplified polygon, brought Dixie to 36 s; running fires in parallel processes brought the whole run to 100 s. Separately, 21 GOES files carried time units that xarray refuses to decode; the scan now reads them with `decode_times=False`, since the timestamp is in the file name.

4 How independent are the sources?

4.1 Points

The brief asked whether FPA-FOD, CAL FIRE and IRWIN are independent. The answer is that *most of the catalog is one record seen through several windows.* Figure 2 shows the copy relationships detected per fire by the rule of §2.3; Table 3 shows what the compiled sources actually contain.

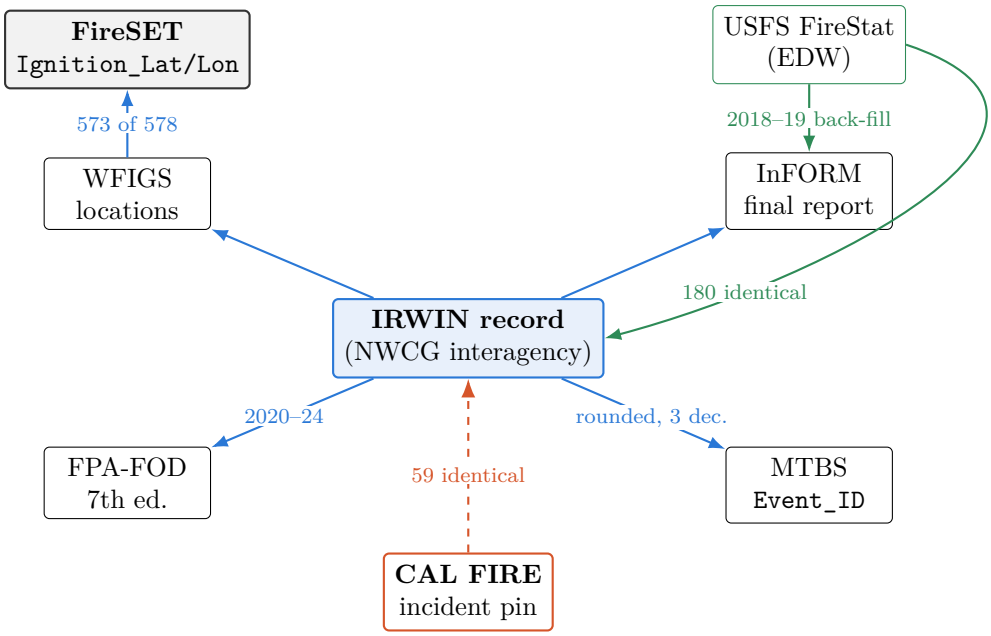


Figure 2: Who copies whom. Solid blue and green arrows: the same point re-published (within 5 m, or a 3–4-decimal rounding); the dashed arrow: an independent record that nevertheless carries the same coordinates on some fires. Counts are fires among the 586 where the two points are identical. FireSET's ignition point is IRWIN's current point on 573 fires; InFORM, FPA-FOD (from 2020) and the MTBS `Event_ID` carry the same point; FireStat equals IRWIN on 180 fires. Only the CAL FIRE incident pin is routinely a separate observation.

Table 3: What the compiled sources actually contain for the 586 fires, by the system each record was copied from.

FPA-FOD 7th ed. (459 matched fires)			InFORM public (545 matched fires)		
`SOURCE_SYSTEM`	lineage	fires	`CreatedBySystem`	lineage	fires
IA-IRWIN, IA-INFORM	IRWIN	340	CAD systems via IRWIN	IRWIN	458
FS-FIRESTAT	USFS-FIRESTAT	56	FireStat	USFS-FIRESTAT	53
DOI-WFMI	DOI-WFMI	35	WFMI	DOI-WFMI	34
IA-ICS209	ICS209	14			
ST-NASF, ST-CACDF	CALFIRE-STATE	10			
FWS-FMIS, IA-MTBS	other	4			

After merging copies, the number of *independent* reporting lineages per fire is small (Table 4): four fires have no agency point, 195 have exactly one (almost always IRWIN), and only 52 have three or more. Where a second lineage exists it is usually the CAL FIRE incident pin, or a FireStat or state record from 2018–19 before those systems were folded into IRWIN. FPA-FOD remains valuable for 2018–19, when its records are FireStat, WFMI and state copies that predate IRWIN, but for 2020 onward it cannot confirm FireSET.

Table 4: Independent reporting lineages per fire, after merging copied points.

Lineages	0	1	2	3	4
Fires	4	195	335	49	3

4.2 Perimeters

Perimeters behave the same way (Figure 3). FRAP, WFIGS, USFS EDW and the NIFC history layer share one polygon on most fires: the intersection-over-union is at least 0.999 on 69–94% of the fires they have in common. MTBS, mapped from Landsat differenced burn ratios rather than by incident teams, is the only independent perimeter. It agrees with the agency polygon at a median IoU of 0.94 and is a median 2% *larger*, not smaller as the catalog expected ("unburned islands are excluded"); unburned islands inside agency polygons are evidently outweighed by burned area outside them.

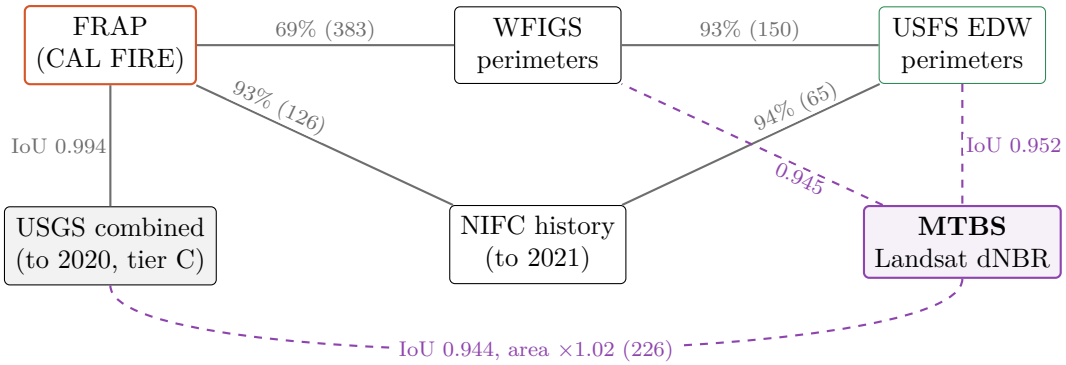


Figure 3: Perimeter agreement between sources. Grey edges: effectively the same polygon passed between systems; the label is the share of shared fires with IoU ≥ 0.999 (number of shared fires in brackets), or the median IoU. Dashed purple edges: MTBS, the one independently mapped perimeter, with its median IoU against each agency layer.

5 Ignition points

5.1 Agreement across independent lineages

Among the 387 fires with two or more independent lineages, 124 agree within 500 m, 154 differ by up to 2 km and 109 differ by more than 2 km (Figure 4). Table 5 grades each source against the consensus of the *other* lineages. The sources are remarkably alike: a median disagreement of 700–900 m and a 90th percentile of 4–5 km, whichever agency is doing the reporting. MTBS's `Event_ID` point is coarser (1.3 km median), as expected for a 3-decimal encoding of a point that is itself a copy. About 5% of WFIGS points carry only two decimals, i.e. roughly ±550 m of precision. Roughly a quarter of agency points sit more than 375 m outside the final perimeter, which for a point of origin is implausible, and a sign that many "ignition" points are dispatch or report locations.

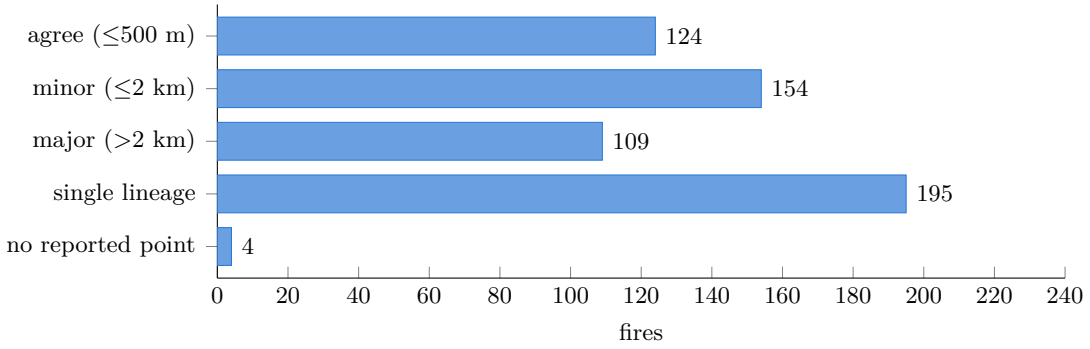


Figure 4: Ignition agreement across independent lineages, 586 fires. "Agree" means every independent agency point lies within 500 m of every other.

Table 5: Distance from each source's ignition point to the consensus of the other independent lineages (leave-one-lineage-out). "Outside" = more than 375 m outside the consensus perimeter.

Source	n	median	90th pct	> 2 km	outside	median to first VIIRS
USFS FireStat (EDW)	139	719 m	4.0 km	13%	9%	634 m
InFORM	348	748 m	4.6 km	15%	17%	790 m
WFIGS / IRWIN	363	793 m	4.6 km	16%	23%	862 m
CAL FIRE incidents	340	805 m	5.0 km	21%	28%	1.20 km
FPA-FOD	280	907 m	4.8 km	17%	18%	793 m
MTBS (`Event_ID`)	143	1.33 km	5.7 km	25%	22%	1.40 km
FireSET (vs all lineages)	578	152 m	1.9 km	10%	24%	873 m

5.2 The first-report point is not the point of origin

IRWIN keeps two locations per incident: `InitialLatitude/Longitude`, entered at first report, and the feature geometry, which is updated as the incident is investigated. They differ by more than 500 m on 193 of 478 fires and by more than 5 km on 56. FireSET correctly uses the current point. Any future comparison that uses the `Initial*` fields (as the delivered catalog did) measures the evolution of IRWIN's own record, not disagreement between agencies.

5.3 FireSET corrections

FireSET's point lies within 100 m of the best-supported agency point on 435 fires and within 500 m on 459 (Figure 5). The remaining 127 fires are listed in `fireset_ignition_patch.csv` with old and new coordinates, the source and confidence of the new point, both points' signed distance to the consensus perimeter and to the first VIIRS overpass, and an *action* decided by the rule in Figure 6.

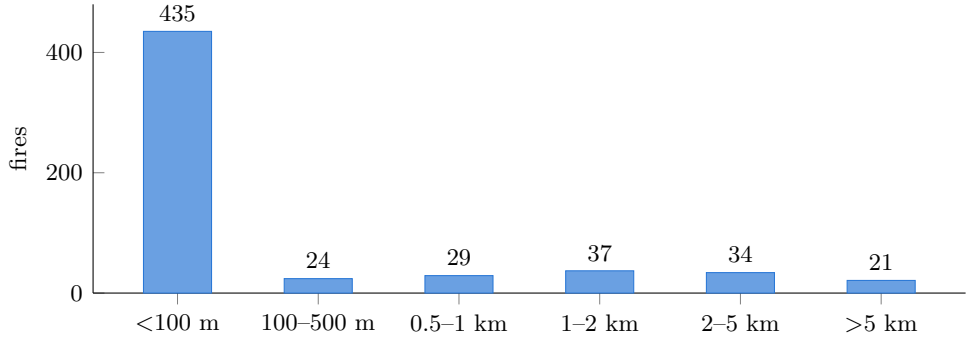


Figure 5: Distance from FireSET's current `Ignition_Lat/Lon` to the recommended agency point, 580 fires with a FireSET point.

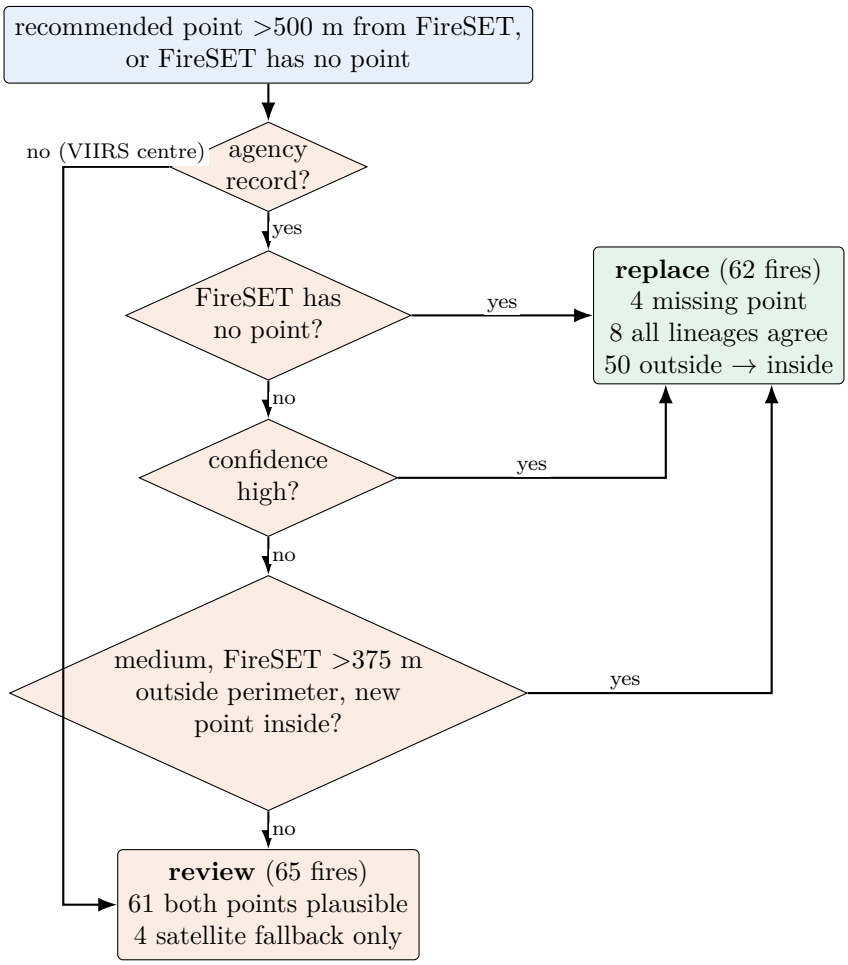


Figure 6: How each proposed correction is classified. "Confidence" is high when all reporting lineages agree, medium when one lineage reports or the winner is strongly supported, low otherwise.

Replace (62 fires). Fifty are cases where FireSET's point lies outside the final perimeter while a second agency point lies inside it; eight are cases where every independent lineage agrees on a point FireSET does not carry; four fill a missing point. The median move is 1.5 km. The replacement points come from the CAL FIRE incident list (31), InFORM final reports (25) and FPA-FOD (6). Two examples:

- **Sites 2024.** FireSET's point is 6.4 km outside the 19,115-acre perimeter. InFORM's final report for the *same* IRWIN incident, and the CAL FIRE pin, agree on a point inside the perimeter 1.1–1.25 km from the first VIIRS overpass. IRWIN's own first-report location is 11 km from its current one: the record was moved once and FireSET captured the wrong state.
- **O'Neals 2018.** FireSET carries a two-decimal point 7.7 km from the 433-acre perimeter. CAL FIRE's pin and the state (ST-NASF) record in FPA-FOD agree within 17 m, 800 m from the perimeter.

Review (65 fires). Both points are plausible (often both inside the perimeter), or the only alternative is the centre of the first VIIRS overpass (4 fires, including 5-6 Cascade Creek 2020, whose FireSET point is 30 km from its 260-acre perimeter and for which no agency record matched at all).

Caveat. Thirty-one replacements rest on the CAL FIRE incident pin. That pin is a map location for the public, not an investigated origin, so rows where it is the only support deserve a look before

they are applied. The scoring also favours points inside the final perimeter, which is right for an origin but is not proof of one.

6 Discovery times

6.1 Agreement and satellite confirmation

IRWIN, CAL FIRE and the state records agree on discovery time: median offset zero, 60–64% within 15 minutes. GOES-R's first fire-mask detection within 5 km of the FireSET point lands a median 0.17 h (10 minutes) after the reported discovery, and within one hour on 275 of the 391 fires where it found anything (Table 6). This is the strongest independent confirmation in the audit: FireSET's `Start_Local` is a genuine report time, not a date-only or back-filled value.

Table 6: Satellite first detection minus reported discovery, hours. VIIRS: first detection inside the perimeter within the lead window; GOES: first fire-mask pixel within 5 km of the FireSET point; HMS: first analyst point within 2 km of the perimeter.

Bound	n	10%	25%	median	75%	90%	< -3 h
GOES-R ABI (5 min)	391	−1.13	−0.06	**0.17**	0.62	2.2	
NOAA HMS (daily)	531	−2.53	−0.18	0.52	2.9	9.2	
VIIRS first overpass	585	−0.97	0.39	3.13	10.6	30	37 fires

VIIRS sees the fire a median 3 h after discovery, as expected for a sensor with about six passes a day. On 37 fires the first VIIRS detection inside the perimeter precedes the reported discovery by more than 3 h: lightning holdovers, late discovery, or heat from a neighbouring fire inside the perimeter. Before the 12-hour lead window was added, Delta 2018 was "detected" 72 h early by the adjacent Hirz Fire.

6.2 A clock error that travels

Table 7 shows discovery-time offsets per source and per record lineage. Two tight clusters stand out, and they are the same error seen from different places (Figure 7).

Table 7: Discovery-time offset from the other lineages' median, hours, for records with a time of day. Counts in the last three columns are records at exactly that offset (±6 min).

Source	record lineage	n	median	within 15 min	at +2 h	at −5 h	at ±7/8 h
WFIGS	IRWIN	435	0.00	60%	1	3	26
InFORM	IRWIN	364	0.00	63%	1	3	24
InFORM	USFS-FIRESTAT	43	**1.97**	5%	**18**	0	0
FPA-FOD	IRWIN	217	−0.03	50%	1	0	0
FPA-FOD	USFS-FIRESTAT	46	**1.98**	4%	**22**	0	0
FPA-FOD	ICS209	11	**2.00**	0%	**9**	0	0
FPA-FOD	DOI-WFMI	29	0.00	55%	0	0	0
USFS EDW points	USFS-FIRESTAT	209	−0.07	33%	0	**22**	**35**
CAL FIRE incidents	CALFIRE-OPS	352	0.05	47%	4	1	2
FireSET		577	0.00	64%	5	2	1

Because a whole-hour offset is symmetric, the audit cannot tell from two clocks alone which one is wrong. The satellite referee resolves it: of the records at suspicious offsets, 67 USFS EDW records are flagged as the erroneous side (2018: 20, 2019: 11, 2020: 36) and none are cleared, whereas 56 of 64 WFIGS records at such offsets are cleared because the satellite sides with IRWIN. CAL FIRE's incident list carries 26 whole-hour errors of its own, 11 of them in 2020. FireSET inherits four.

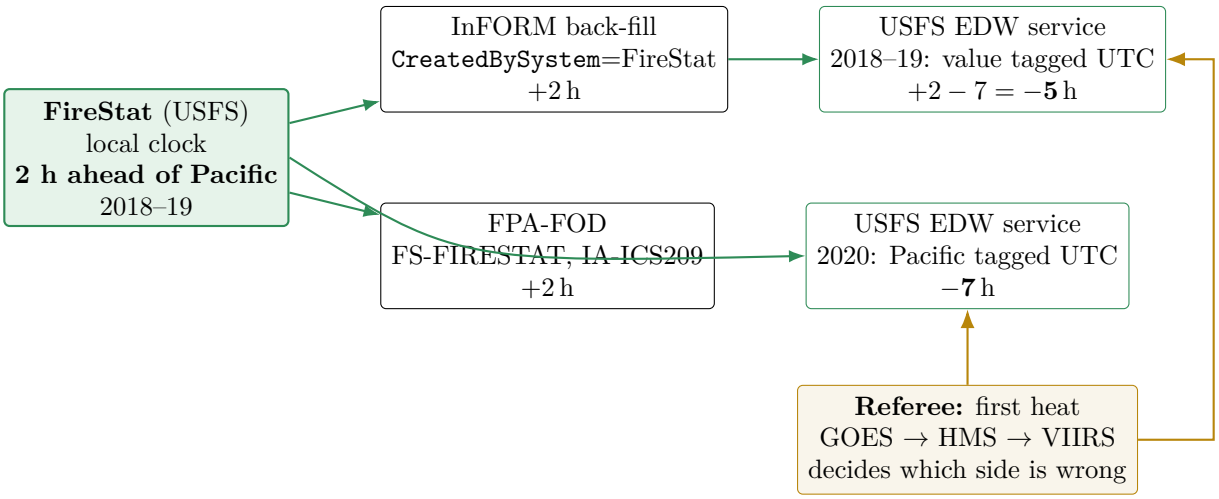


Figure 7: How one clock error appears as three different offsets. FireStat's 2018–19 times run two hours fast (Delta 2018: FireStat 14:51 against 12:51 PDT in CAL FIRE and IRWIN); the copies in InFORM and FPA-FOD inherit +2 h; the EDW map service labels FireStat's local value as UTC, which shows as −5 h in 2018–19 (Delta is served as 19:51Z) and −7 h in 2020. 67 EDW records from 2018–2020 carry the error; EDW records from 2023 on agree with IRWIN. The original catalog looked only for 7- and 8-hour offsets.

7 Size and cause

Reported final sizes agree with the consensus perimeter area to within 0.1% at the median, because they are mostly the same polygon's acreage. 114 fires nevertheless have at least one source more than 25% off. The main cause is IRWIN incident sizes never updated past discovery: 36 fires whose WFIGS `IncidentSize` is 0.01–0.3 acres against hundreds or thousands of acres mapped, an error that InFORM and FPA-FOD inherit. A second cause was a field choice: WFIGS perimeters' `poly_GISAcres` is a placeholder (0.1, 1 or 20 acres) on many 2020 polygons, so the run uses `poly_Acres_AutoCalc`. MTBS's burned area is a median 1–2% larger than the agency figure.

Cause is reported by two or more sources for 570 fires and conflicts between human and natural on only three. The consensus is Undetermined for 279 fires, Natural for 179 and Human for 128; the large Undetermined share is genuine (IRWIN and FRAP both record "undetermined" or code 14), not a classification failure.

8 Insights

1. **There is one upstream record, not many sources.** IRWIN is the system of record for wildfire incidents since 2020, and WFIGS, InFORM, FPA-FOD and MTBS re-publish it. Counting them as corroboration inflates confidence without adding information. FireSET's choice of WFIGS is the right one, but it means FireSET cannot be confirmed by the "federal" sources at all; only CAL FIRE's operational record, pre-2020 FireStat and state reports, and the satellites are independent.
2. **A map pin is not a point of origin.** About a quarter of agency ignition points lie more than 375 m outside the final perimeter. IRWIN's first-report location moves by more than 500 m on 40% of fires before it settles. The honest precision of an "ignition point" in these systems is hundreds of metres, and for a minority, kilometres.
3. **Timing is the strongest-confirmed attribute.** GOES-R puts first heat within an hour of FireSET's start time on 70% of fires it sees; IRWIN, CAL FIRE and the state records agree with each other to the minute. The errors that do exist are systematic clock errors in one system (FireStat), not noise, and can be corrected by rule.
4. **Perimeter and size are a single measurement.** FRAP's polygon is *the* polygon; the other

agency layers copy it and its acreage. MTBS is the one check, and it agrees to IoU 0.94 while being slightly larger.

5. **Validating a dataset against live services is not the same as reading their metadata.** Seven field-level defects (§3) survived a metadata check. Each would have produced a confident, wrong conclusion about FireSET.

9 Recommendations for FireSET

1. Apply the 62 *replace* rows of `fireset_ignition_patch.csv` after a spot check of the 31 that rest on a CAL FIRE pin alone; keep the provenance column so each coordinate traces to an agency record.
2. Review the 65 *review* rows in QGIS. The layers are `ignition_points`, `consensus_perimeters` and `source_perimeters`; most rows are two plausible points inside one perimeter.
3. Record an *ignition precision* field: the number of decimals carried, whether the point lies inside the final perimeter, and whether a second independent lineage exists. For 195 fires no second lineage exists.
4. Keep `Start_Local`; it is confirmed by GOES-R. Where a USFS source is used for timing, correct the 2018–20 FireStat/EDW offsets by rule rather than by hand.
5. Treat FRAP and MTBS as the two perimeter measurements; ignore WFIGS/EDW/NIFC perimeters as duplicates.

10 Limits of this audit

- The GOES search is centred on FireSET's own point (5 km radius), so it confirms timing but is not an independent *location* check. FireSET's own VIIRS `.h5` files were not available; the FIRMS S-NPP and NOAA-20 yearly archives were used instead (no NOAA-21), with "type 0" detections only.
- The recommended point is always an existing agency record scored on support, perimeter, satellite proximity, trust and precision; the audit never synthesises coordinates except as a low-confidence VIIRS fallback for fires with no agency point.
- NASA FEDS perimeters, InciWeb and NIROPS infrared perimeters were not used (no archive endpoint, or manual sources); the CAL FIRE FIRIS view holds only the current season.
- Thresholds (500 m / 2 km agreement, 375 m perimeter tolerance, 12 h lead window, IoU 0.7) are settings in `settings.yaml`, not truths.

A Reproducing the run

```
git clone https://github.com/utkrshkmr/fire-data-revalid
cd fire-data-revalid
scripts/download_local_sources.sh          # FPA-FOD, MTBS, USGS, FIRMS (~2.5 GB)
cd fire-source-audit
python -m venv .venv && .venv/bin/pip install -e ".[dev,goes]"
cp /path/to/SUBSET_expanded_20260830.xlsx data/
.venv/bin/pytest -q                        # 24 tests
.venv/bin/fsa probe                        # every endpoint and field mapping
.venv/bin/fsa fetch --workers 6            # network sources -> cache/
.venv/bin/fsa --only viirs_firms_csv,hms_points fetch
.venv/bin/fsa --only goes_fdc fetch        # optional, ~20 min on 16 cores
.venv/bin/fsa analyze                      # -> out/
python ../scripts/summarize_findings.py out   # the numbers in this document
```

B Output files

File	Contents
`fire_summary.csv`	one row per fire: consensus, agreement class, recommendation, satellite bounds, flags
`source_errors.csv`	one row per fire × source × aspect, with every metric
`source_summary.csv`	per-source statistics
`timing_by_lineage.csv`	discovery offsets per source and record lineage
`identical_point_pairs.csv`	every detected copy of a point between sources, by lineage
`perimeter_pairs.csv`	IoU, area ratio and Hausdorff distance for every pair of perimeters
`flags.csv`	every issue, with category and severity
`fireset_ignition_patch.csv`	the 127 proposed corrections with action, evidence and provenance
`matched_records.csv`	the crosswalk: which record in each source matched each fire, and how
`ignition_points.geojson`	reported, consensus, recommended and first-VIIRS points per fire
`consensus_perimeters.geojson`, `source_perimeters.geojson`	perimeter layers for QGIS
`report.html`, `fire_source_audit.xlsx`	HTML summary with charts; the main tables in one workbook